

- 1 An electric heater is to be made from nichrome wire. Nichrome has a resistivity of $1.0 \times 10^{-6} \Omega \text{m}$ at the operating temperature of the heater. The heater is to have a power dissipation of 60W when the potential difference across its terminals is 12V.

(a) For the heater operating at its designed power,

(i) calculate the current,

current = A [2]

(ii) show that the resistance of the nichrome wire is 2.4Ω .

[2]

(b) Calculate the length of nichrome wire of diameter 0.80 mm required for the heater.

length = m [3]

- (c) A second heater, also designed to operate from a 12V supply, is constructed using the same nichrome wire but using half the length of that calculated in (b). Explain quantitatively the effect of this change in length of wire on the power of the heater.

.....

 [3]

2 (a) Define the *ohm*.

..... [1]

(b) Determine the SI base units of **resistivity**.

base units of resistivity = [3]

(c) A cell of e.m.f. 2.0V and negligible internal resistance is connected to a variable resistor R and a metal wire, as shown in Fig. 5.1.

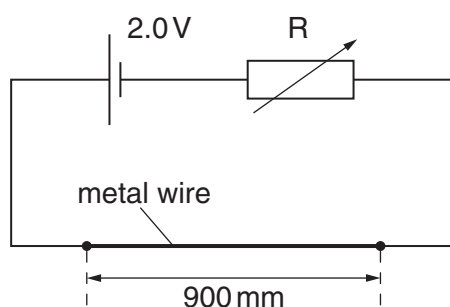


Fig. 5.1

The wire is 900mm long and has an area of cross-section of $1.3 \times 10^{-7} \text{ m}^2$. The resistance of the wire is 3.4Ω .

(i) Calculate the resistivity of the metal wire.

resistivity = [2]

- (ii) The resistance of R may be varied between 0 and $1500\ \Omega$.
Calculate the maximum potential difference (p.d.) and minimum p.d. possible across the wire.

maximum p.d. = V

minimum p.d. =V
[2]

- (iii) Calculate the power transformed in the wire when the potential difference across the wire is 2.0V.

power = W [2]

- (d) Resistance R in (c) is now replaced with a different variable resistor Q. State the power transformed in Q, for Q having

- (i) zero resistance,

power = W [1]

- (ii) infinite resistance.

power = W [1]

3 (a) The output of a heater is 2.5 kW when connected to a 220 V supply. 9702/21/M/J/12/Q4

(i) Calculate the resistance of the heater.

resistance = Ω [2]

(ii) The heater is made from a wire of cross-sectional area $2.0 \times 10^{-7} \text{ m}^2$ and resistivity $1.1 \times 10^{-6} \Omega \text{ m}$.

Use your answer in (i) to calculate the length of the wire.

length = m [3]

(b) The supply voltage is changed to 110 V.

(i) Calculate the power output of the heater at this voltage, assuming there is no change in the resistance of the wire.

power = W [1]

(ii) State and explain quantitatively **one** way that the wire of the heater could be changed to give the same power as in (a).

.....
.....
..... [2]

- 4 Fig. 5.1 shows a 12 V power supply with negligible internal resistance connected to a uniform metal wire AB. The wire has length 1.00 m and resistance $10\ \Omega$. Two resistors of resistance $4.0\ \Omega$ and $2.0\ \Omega$ are connected in series across the wire.

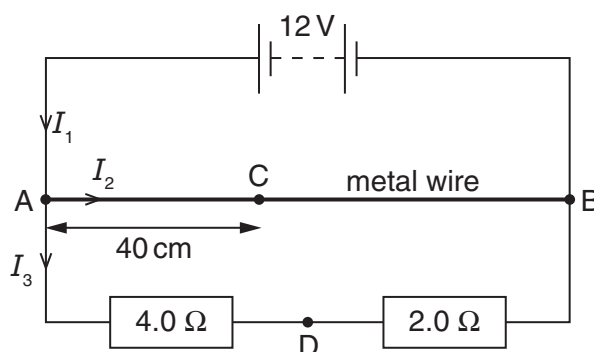


Fig. 5.1

Currents I_1 , I_2 and I_3 in the circuit are as shown in Fig. 5.1.

- (a) (i) Use Kirchhoff's first law to state a relationship between I_1 , I_2 and I_3 .

..... [1]

- (ii) Calculate I_1 .

$I_1 =$ A [3]

- (iii) Calculate the ratio x , where

$$x = \frac{\text{power in metal wire}}{\text{power in series resistors}}.$$

$x =$ [3]

- (b) Calculate the potential difference (p.d.) between the points C and D, as shown in Fig. 5.1. The distance AC is 40 cm and D is the point between the two series resistors.

p.d. = V [3]

- 5 (a) Define electrical *resistance*.

.....
 [1]

- (b) A circuit is set up to measure the resistance R of a metal wire. The potential difference (p.d.) V across the wire and the current I in the wire are to be measured.

- (i) Draw a circuit diagram of the apparatus that could be used to make these measurements.

[3]

- (ii) Readings for p.d. V and the corresponding current I are obtained. These are shown in Fig. 2.1.

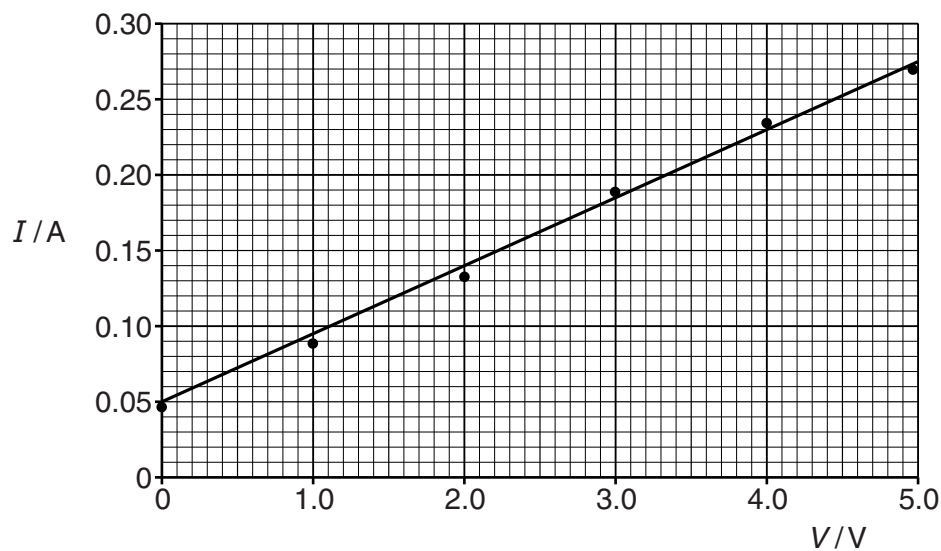


Fig. 2.1

Explain how Fig. 2.1 indicates that the readings are subject to

1. a systematic uncertainty,

.....
..... [1]

2. random uncertainties.

.....
..... [1]

(iii) Use data from Fig. 2.1 to determine R . Explain your working.

$R = \dots\dots\dots \Omega$ [3]

(c) In another experiment, a value of R is determined from the following data:

Current $I = 0.64 \pm 0.01 \text{ A}$ and p.d. $V = 6.8 \pm 0.1 \text{ V}$.

Calculate the value of R , together with its uncertainty. Give your answer to an appropriate number of significant figures.

$R = \dots\dots\dots \pm \dots\dots\dots \Omega$ [3]

- 6 (a) Define *potential difference* (p.d.).

..... [1]

- (b) A power supply of e.m.f. 240 V and zero internal resistance is connected to a heater as shown in Fig. 6.1.



Fig. 6.1

The wires used to connect the heater to the power supply each have length 75 m. The wires have a cross-sectional area 2.5 mm^2 and resistivity $18 \text{ n}\Omega \text{ m}$. The heater has a constant resistance of 38Ω .

- (i) Show that the resistance of each wire is 0.54Ω .

[3]

- (ii) Calculate the current in the wires.

current = A [3]

- (iii) Calculate the power loss in the wires.

power = W [3]

- (c) The wires to the heater are replaced by wires of the same length and material but having a cross-sectional area of 0.50 mm^2 . Without further calculation, state and explain the effect on the power loss in the wires.

.....

..... [2]

- 7 A battery connected in series with a resistor R of resistance 5.0Ω is shown in Fig. 6.1.

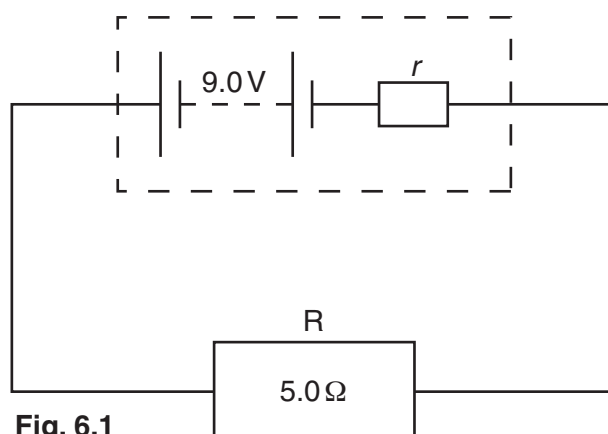


Fig. 6.1

The electromotive force (e.m.f.) of the battery is 9.0V and the internal resistance is r .
The potential difference (p.d.) across the battery terminals is 6.9V .

- (a) Use energy considerations to explain why the p.d. across the battery is not equal to the e.m.f. of the battery.

.....
 [2]

- (b) Calculate

- (i) the current in the circuit,

current = A [2]

- (ii) the internal resistance r .

$r = \dots\dots\dots \Omega$ [2]

- (c) Calculate, for the battery in the circuit,

- (i) the total power produced,

power = W [2]

- (ii) the efficiency.

efficiency = [2]

- 8 (a) Distinguish between *electromotive force* (e.m.f.) and *potential difference* (p.d.).

.....

 [2]

- (b) A battery of e.m.f. 12V and internal resistance $0.50\ \Omega$ is connected to two identical lamps, as shown in Fig. 6.1.

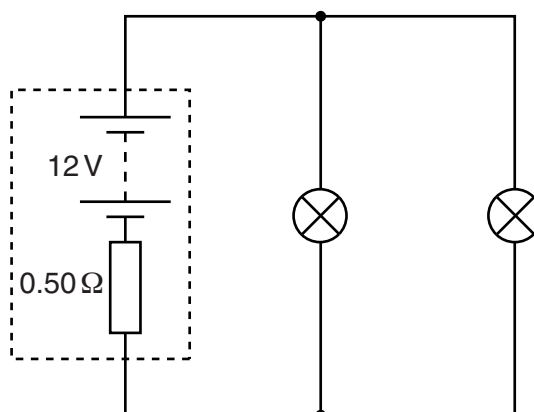


Fig. 6.1

Each lamp has constant resistance. The power rating of each lamp is 48W when connected across a p.d. of 12V.

- (i) Explain why the power dissipated in each lamp is not 48W when connected as shown in Fig. 6.1.

.....

 [1]

- (ii) Calculate the resistance of one lamp.

resistance = Ω [2]

- (iii) Calculate the current in the battery.

current = A [2]

(iv) Calculate the power dissipated in one lamp.

power = W [2]

(c) A third identical lamp is placed in parallel with the battery in the circuit of Fig. 6.1. Describe and explain the effect on the terminal p.d. of the battery.

.....

 [2]

9702/23/M/J/14/Q5

9 (a) Explain why the terminal potential difference (p.d.) of a cell with internal resistance may be less than the electromotive force (e.m.f.) of the cell.

.....

 [2]

(b) A battery of e.m.f. 4.5V and internal resistance r is connected in series with a resistor of resistance 6.0Ω , as shown in Fig. 5.1.

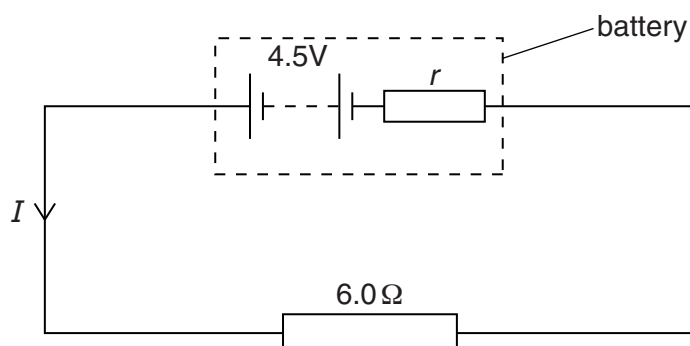


Fig. 5.1

The current I in the circuit is 0.65 A.

Determine

(i) the internal resistance r of the battery,

$r = \dots\dots\dots \Omega$ [2]

(ii) the terminal p.d. of the battery,

p.d. = $\dots\dots\dots$ V [2]

(iii) the power dissipated in the resistor,

power = $\dots\dots\dots$ W [2]

(iv) the efficiency of the battery.

efficiency = $\dots\dots\dots$ [2]

(c) A second resistor of resistance 20Ω is connected in parallel with the 6.0Ω resistor in Fig. 5.1.

Describe and explain qualitatively the change in the heating effect within the battery.

.....
.....
.....
.....[3]

- 10 The resistance R of a uniform metal wire is measured for different lengths l of the wire. The variation with l of R is shown in Fig. 3.1.

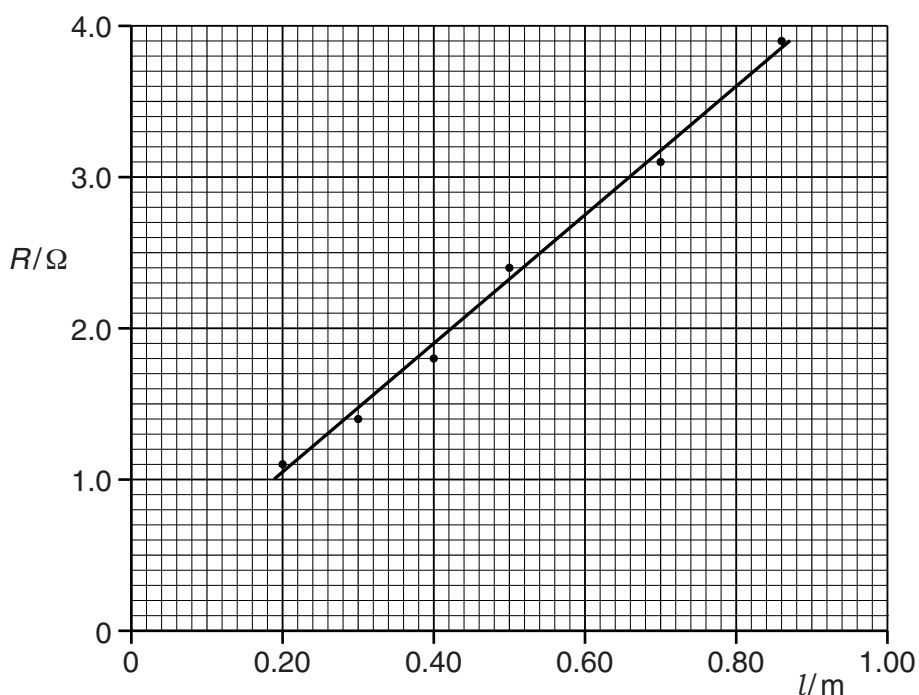


Fig. 3.1

- (a) The points shown in Fig. 3.1 do not lie on the best-fit line. Suggest a reason for this.

.....
 [1]

- (b) Determine the gradient of the line shown in Fig. 3.1.

gradient = [2]

- (c) The cross-sectional area of the wire is 0.12 mm^2 .

Use your answer in (b) to determine the resistivity of the metal of the wire.

resistivity = $\Omega \text{ m}$ [3]

- (d) The resistance R of different wires is measured. The wires are of the same metal and same length but have different cross-sectional areas A .

On Fig. 3.2, sketch a graph to show the variation with A of R .



Fig. 3.2

- 11 The variation with potential difference (p.d.) V of current I for a semiconductor diode is shown in Fig. 5.1.

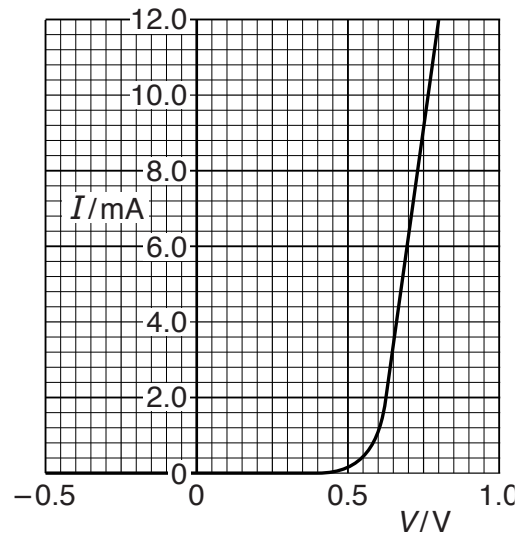


Fig. 5.1

- (a) Use Fig. 5.1 to describe the variation of the resistance of the diode between $V = -0.5$ V and $V = 0.8$ V.

.....
[2]

- (b) On Fig. 5.2, sketch the variation with p.d. V of current I for a filament lamp. Numerical values are not required.

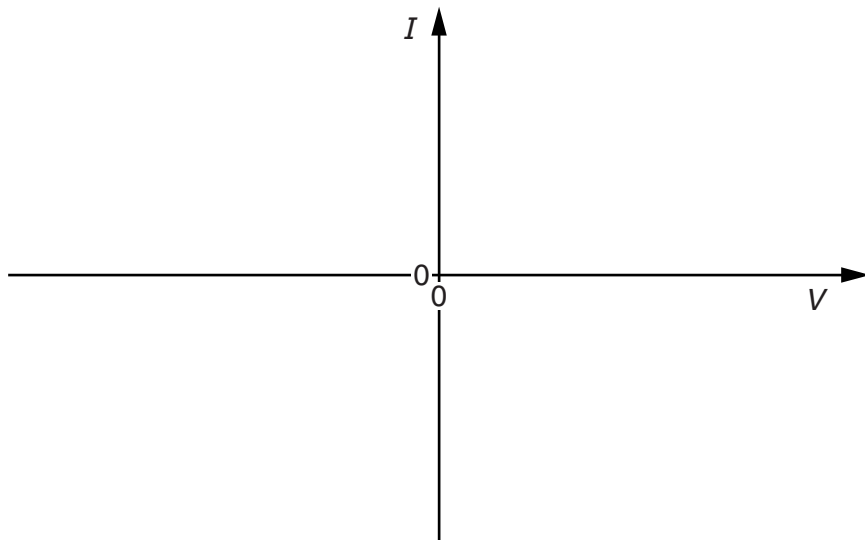


Fig. 5.2

[2]

- (c) Fig. 5.3 shows a power supply of electromotive force (e.m.f.) 12V and internal resistance 0.50Ω connected to a filament lamp and switch.

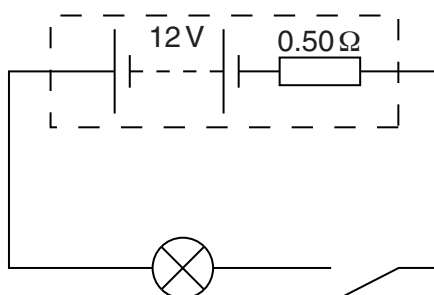


Fig. 5.3

The filament lamp has a power of 36W when the p.d. across it is 12V.

- (i) Calculate the resistance of the lamp when the p.d. across it is 12V.

resistance = Ω [1]

- (ii) The switch is closed and the current in the lamp is 2.8 A. Calculate the resistance of the lamp.

resistance = Ω [3]

- (d) Explain how the two values of resistance calculated in (c) provide evidence for the shape of the sketch you have drawn in (b).

.....
[1]

12 (a) (i) State what is meant by an *electric current*.

9702/22/F/M/16/Q5

.....
..... [1]

(ii) Define electric *potential difference* (*p.d.*).

.....
..... [1]

(b) A power supply of electromotive force (e.m.f.) 8.7 V and negligible internal resistance is connected by two identical wires to three filament lamps, as shown in Fig. 5.1.

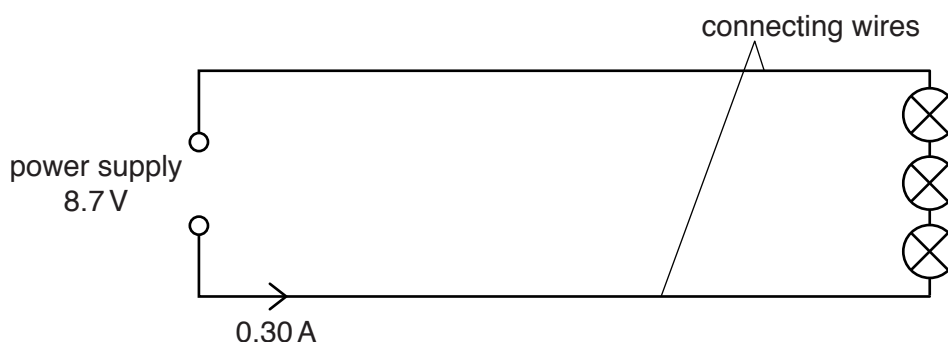


Fig. 5.1 (not to scale)

The power supply provides a current of 0.30 A to the circuit.
The filament lamps are identical. The I - V characteristic for **one** of the lamps is shown in Fig. 5.2.

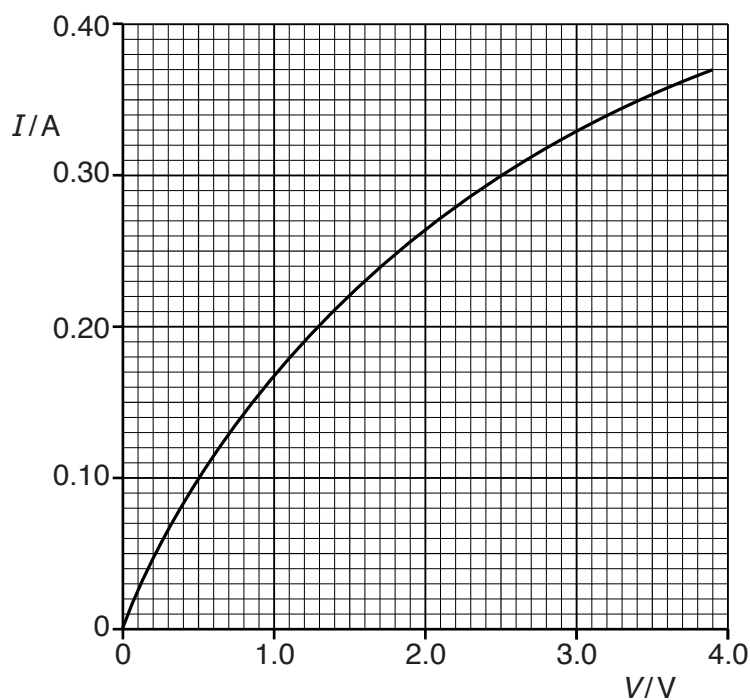


Fig. 5.2

- (i) Show that the resistance of each connecting wire is $2.0\,\Omega$.

[2]

- (ii) The resistivity of the metal of the connecting wires does not vary with temperature. On Fig. 5.2, sketch the I – V characteristic for **one** of the connecting wires.

[2]

- (iii) Calculate the power loss in one of the connecting wires.

power = W [2]

- (iv) Some data for the connecting wires are given below.

cross-sectional area = 0.40 mm^2

resistivity = $1.7 \times 10^{-8}\,\Omega\text{ m}$

number density of free electrons = $8.5 \times 10^{28}\text{ m}^{-3}$

Calculate

1. the length of one of the connecting wires,

length = m [2]

2. the drift speed of a free electron in the connecting wires.

drift speed = ms^{-1} [2]

13 (a) Define the *coulomb*.

9702/21/M/J/16/Q6

.....[1]

(b) A resistor X is connected to a cell as shown in Fig. 6.1.

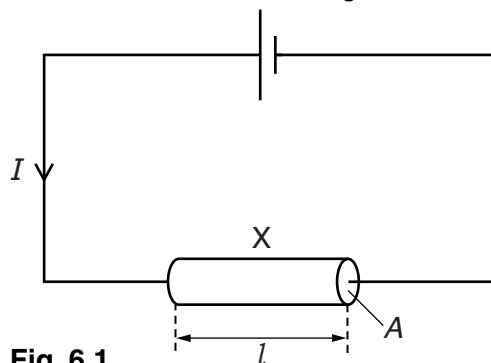


Fig. 6.1

The resistor is a wire of cross-sectional area A and length l . The current in the wire is I .

Show that the average drift speed v of the charge carriers in X is given by the equation

$$v = \frac{I}{nAe}$$

where e is the charge on a charge carrier and n is the number of charge carriers per unit volume in X .

[3]

(c) A 12V battery with negligible internal resistance is connected to two resistors Y and Z, as shown in Fig. 6.2.

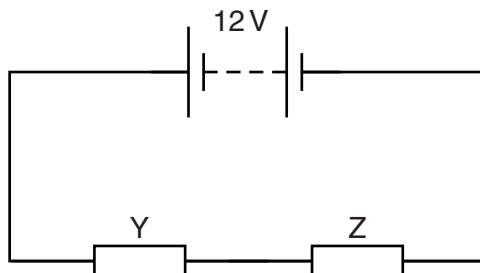


Fig. 6.2

The resistors are made from wires of the same material. The wire of Y has a diameter d and length l . The wire of Z has a diameter $2d$ and length $2l$.

(i) Determine the ratio

$$\frac{\text{average drift speed of the charge carriers in Y}}{\text{average drift speed of the charge carriers in Z}} .$$

ratio = [3]

(ii) Show that

$$\frac{\text{resistance of Y}}{\text{resistance of Z}} = 2.$$

[2]

(iii) Determine the potential difference across Y.

potential difference = V [2]

(iv) Determine the ratio

$$\frac{\text{power dissipated in Y}}{\text{power dissipated in Z}} .$$

ratio = [1]

- 14 (a) Electric current is a flow of charge carriers. The charge on the carriers is quantised. Explain what is meant by *quantised*.

.....[1]

- (b) A battery of electromotive force (e.m.f.) 9.0 V and internal resistance $0.25\ \Omega$ is connected in series with two identical resistors X and a resistor Y, as shown in Fig. 7.1.

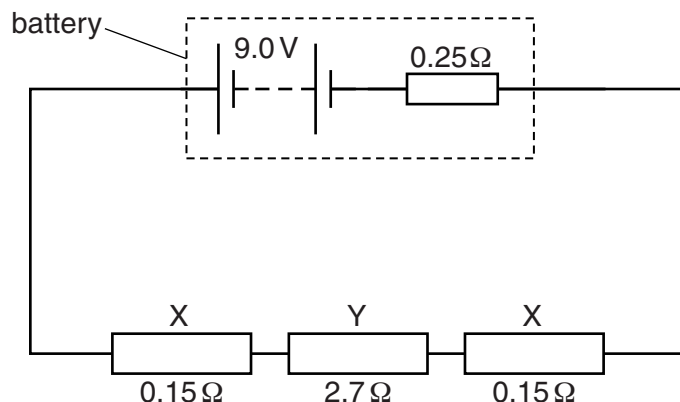


Fig. 7.1

The resistance of each resistor X is $0.15\ \Omega$ and the resistance of resistor Y is $2.7\ \Omega$.

- (i) Show that the current in the circuit is 2.8 A .

[3]

- (ii) Calculate the potential difference across the battery.

potential difference = V [2]

- (c) Each resistor X connected in the circuit in (b) is made from a wire with a cross-sectional area of 2.5 mm^2 . The number of free electrons per unit volume in the wire is $8.5 \times 10^{29} \text{ m}^{-3}$.

(i) Calculate the average drift speed of the electrons in X.

drift speed = ms^{-1} [2]

- (ii) The two resistors X are replaced by two resistors Z made of the same material and length but with half the diameter.

Describe and explain the difference between the average drift speed in Z and that in X.

.....

 [2]

9702/23/M/J/16/Q6

- 15 (a) Define the *ohm*.

..... [1]

- (b) A 15V battery with negligible internal resistance is connected to two resistors P and Q, as shown in Fig. 6.1.

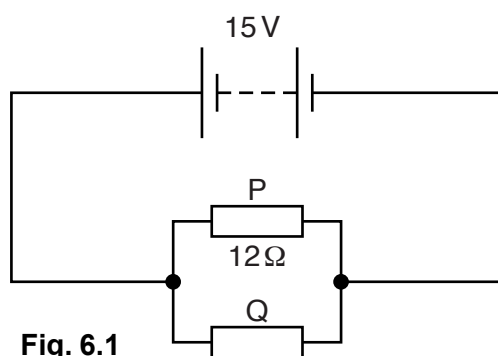


Fig. 6.1

The resistors are made of wires of the same material. The wire of P has diameter d and length $2l$. The wire of Q has diameter $2d$ and length l .

The resistance of P is $12\ \Omega$.

(i) Show that the resistance of Q is $1.5\ \Omega$.

(ii) Calculate the total power dissipated in the resistors P and Q.

[3]

power = W [3]

(iii) Determine the ratio

$$\frac{\text{average drift speed of the charge carriers in P}}{\text{average drift speed of the charge carriers in Q}}.$$

ratio = [3]

- 16 A wire of cross-sectional area 1.5 mm^2 and length 2.5 m has a resistance of 0.030Ω .
Calculate the resistivity of the material of the wire in $\text{n}\Omega \text{ m}$.

resistivity = $\text{n}\Omega \text{ m}$ [3]

17 (a) Define the *ohm*.

.....[1]

(b) Wires are used to connect a battery of negligible internal resistance to a lamp, as shown in Fig. 7.1.

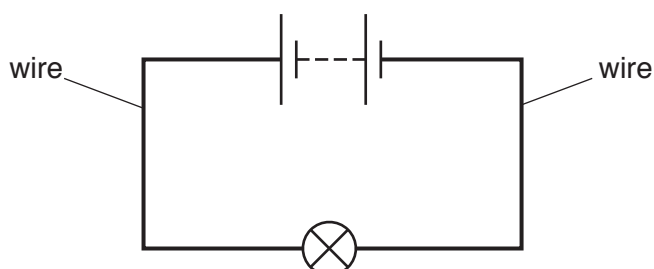


Fig. 7.1

The lamp is at its normal operating temperature. Some data for the filament wire of the lamp and for the connecting wires of the circuit are shown in Fig. 7.2.

	filament wire	connecting wires
diameter	d	$14d$
total length	L	$7.0L$
resistivity of metal (at normal operating temperature)	ρ	0.028ρ

Fig. 7.2

(i) Show that

$$\frac{\text{resistance of filament wire}}{\text{total resistance of connecting wires}} = 1000.$$

- (ii) Use the information in (i) to explain qualitatively why the power dissipated in the filament wire of the lamp is greater than the total power dissipated in the connecting wires.

.....
.....
.....[1]

- (iii) The lamp is rated as 12V, 6.0W. Use the information in (i) to determine the total resistance of the connecting wires.

total resistance of connecting wires = Ω [3]

- (iv) The diameter of the connecting wires is decreased. The total length of the connecting wires and the resistivity of the metal of the connecting wires remain the same.

State and explain the change, if any, that occurs to the resistance of the filament wire of the lamp.

.....
.....
.....
.....
.....[3]

18 (a) State what is meant by an *electric current*.

9702/22/O/N/17/Q6

.....[1]

(b) A metal wire has length L and cross-sectional area A , as shown in Fig. 6.1.

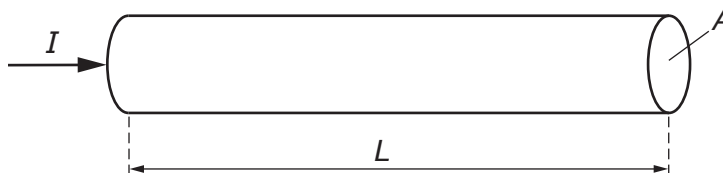


Fig. 6.1

I is the current in the wire,

n is the number of free electrons per unit volume in the wire,

v is the average drift speed of a free electron and

e is the charge on an electron.

(i) State, in terms of A , e , L and n , an expression for the total charge of the free electrons in the wire.

.....[1]

(ii) Use your answer in (i) to show that the current I is given by the equation

$$I = nAve.$$

[2]

(c) A metal wire in a circuit is damaged. The resistivity of the metal is unchanged but the cross-sectional area of the wire is reduced over a length of 3.0 mm, as shown in Fig. 6.2.

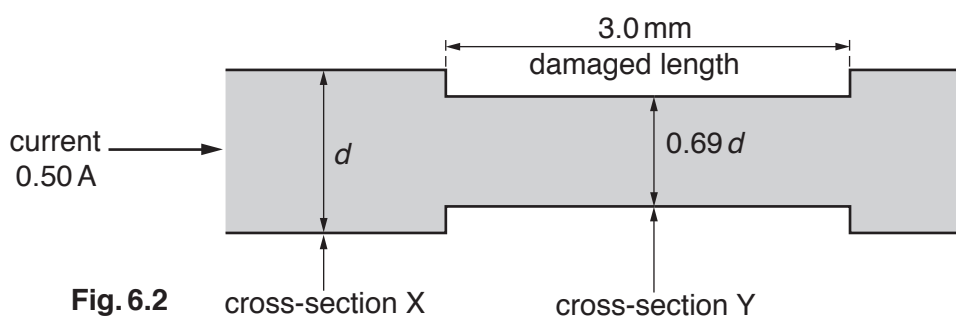


Fig. 6.2

The wire has diameter d at cross-section X and diameter $0.69d$ at cross-section Y.
The current in the wire is 0.50 A.

(i) Determine the ratio

$$\frac{\text{average drift speed of free electrons at cross-section Y}}{\text{average drift speed of free electrons at cross-section X}}$$

ratio = [2]

(ii) The main part of the wire with cross-section X has a resistance per unit length of $1.7 \times 10^{-2} \Omega \text{ m}^{-1}$.

For the damaged length of the wire, calculate

1. the resistance per unit length,

resistance per unit length = $\Omega \text{ m}^{-1}$ [2]

2. the power dissipated.

power = W [2]

(iii) The diameter of the damaged length of the wire is further decreased. Assume that the current in the wire remains constant.

State and explain qualitatively the change, if any, to the power dissipated in the damaged length of the wire.

.....
.....
..... [2]

19 A filament lamp is rated as 30 W, 120 V. A potential difference of 120 V is applied across the lamp.

(a) For the filament wire of the lamp, calculate

(i) the current,

current = A [2]

(ii) the number of electrons passing a point in 3.0 hours.

number = [2]

(b) Show that the resistance of the filament wire is $480\ \Omega$.

[2]

(c) The filament wire has an uncoiled length of 580 mm and is made of metal. The metal has resistivity $6.1 \times 10^{-7}\ \Omega\text{ m}$ at the operating temperature of the lamp.

Calculate the diameter of the wire.

diameter = m [3]

(d) The potential difference across the lamp is now reduced. State and explain the effect, if any, on the resistance of the filament wire.

.....

..... [1]

20 (a) Define the *volt*.

9702/21/M/J/18/Q6

.....[1]

- (b) A battery of electromotive force (e.m.f.) 4.5 V and negligible internal resistance is connected to two filament lamps P and Q and a resistor R, as shown in Fig. 6.1.

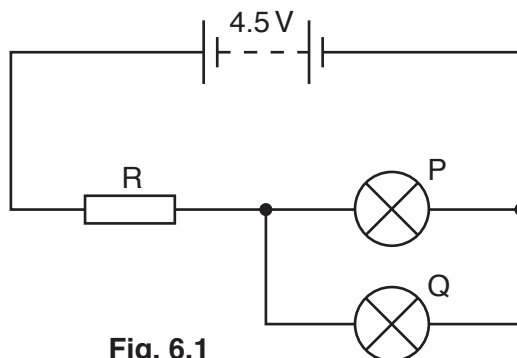


Fig. 6.1

The current in lamp P is 0.15 A.

The I - V characteristics of the filament lamps are shown in Fig. 6.2.

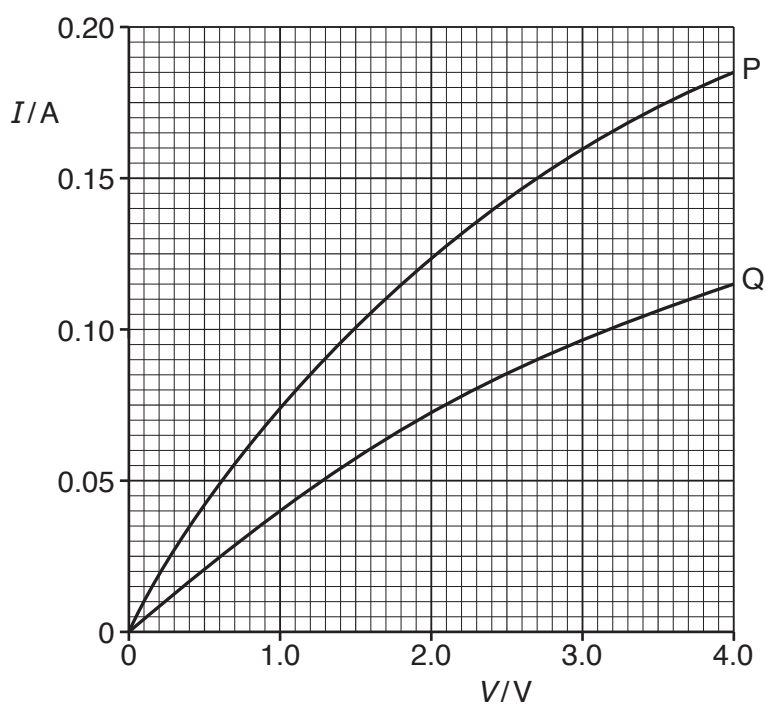


Fig. 6.2

- (i) Use Fig. 6.2 to determine the current in the battery. Explain your working.

current = A [2]

- (ii) Calculate the resistance of resistor R.

resistance = Ω [2]

- (iii) The filament wires of the two lamps are made from material with the same resistivity at their operating temperature in the circuit. The diameter of the wire of lamp P is twice the diameter of the wire of lamp Q.

Determine the ratio

$$\frac{\text{length of filament wire of lamp P}}{\text{length of filament wire of lamp Q}}.$$

ratio = [3]

- (iv) The filament wire of lamp Q breaks and stops conducting.

State and explain, qualitatively, the effect on the resistance of lamp P.

.....
.....
.....
.....[2]

21 A wire X has a constant resistance per unit length of $3.0 \Omega \text{ m}^{-1}$ and a diameter of 0.48 mm .

(a) Calculate the resistivity of the metal of wire X.

resistivity = $\Omega \text{ m}$ [3]

(b) The wire X is connected into the circuit shown in Fig. 6.1.

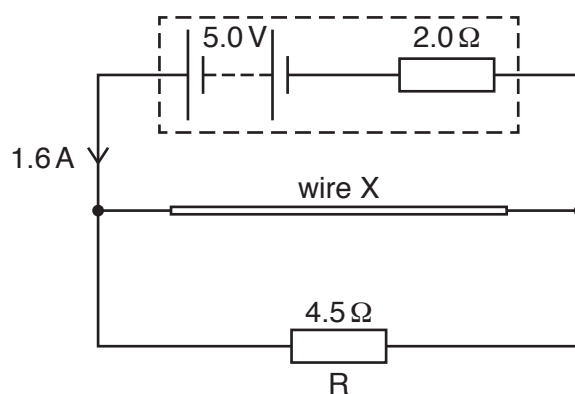


Fig. 6.1

The battery has an electromotive force (e.m.f.) of 5.0 V and an internal resistance of 2.0Ω . The wire X and a resistor R of resistance 4.5Ω are connected in parallel. The current in the battery is 1.6 A .

(i) Calculate the potential difference across resistor R.

potential difference = V [1]

(ii) Determine, for wire X,

1. its resistance,

resistance = Ω [3]

2. its length.

length = m [1]

- 22 (a) The current I in a metal wire is given by the expression

$$I = Anve.$$

State what is meant by the symbols A and n .

A :

n : [2]

- (b) The diameter of a wire XY varies linearly with distance along the wire as shown in Fig. 7.1.

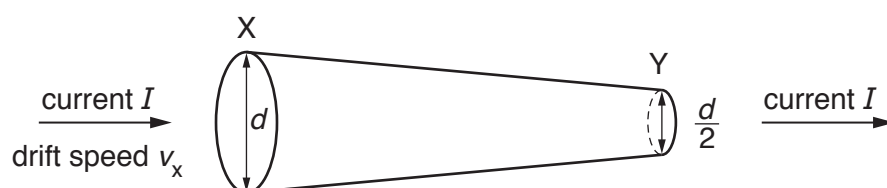


Fig. 7.1

There is a current I in the wire. At end X of the wire, the diameter is d and the average drift speed of the free electrons is v_x . At end Y of the wire, the diameter is $\frac{d}{2}$.

On Fig. 7.2, sketch a graph to show the variation of the average drift speed with position along the wire between X and Y.

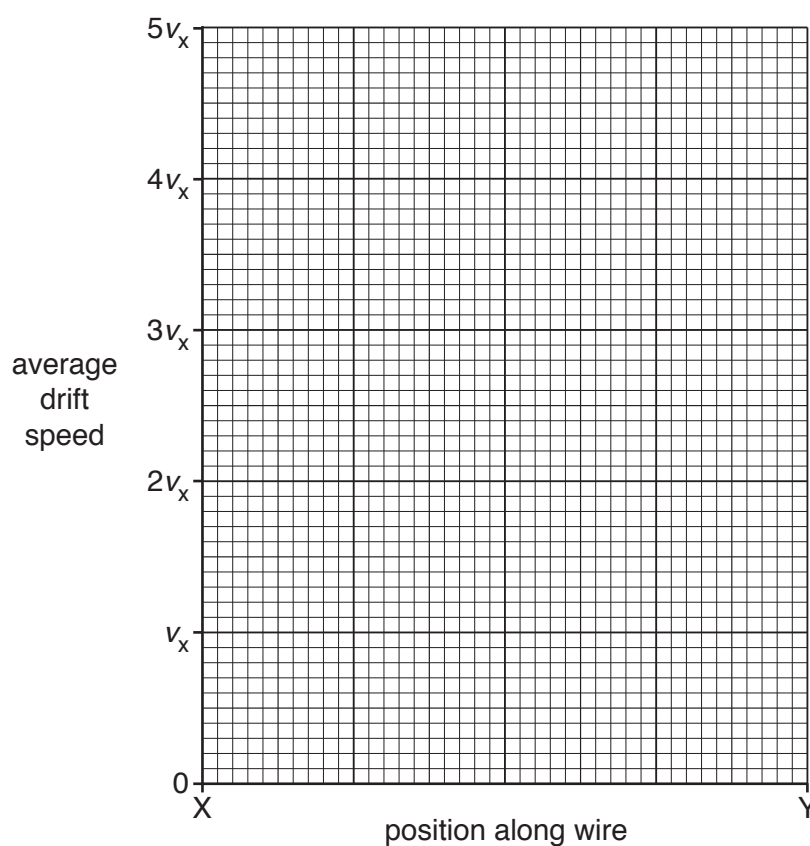


Fig. 7.2

[2]

23 (a) Define the *coulomb*.

.....
[1]

(b) An electric current is a flow of charge carriers.

In the following list, underline the possible charges for a charge carrier.

$8.0 \times 10^{-19} \text{ C}$ $4.0 \times 10^{-19} \text{ C}$ $1.6 \times 10^{-19} \text{ C}$ $1.6 \times 10^{-20} \text{ C}$ [1]

(c) The diameter of a wire ST varies linearly with distance along the wire as shown in Fig. 6.1.

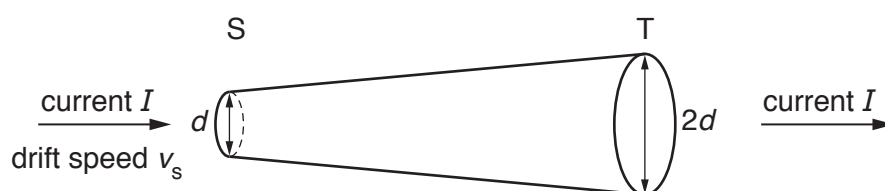


Fig. 6.1

There is a current I in the wire. At end S of the wire, the diameter is d and the average drift speed of the free electrons is v_s . At end T of the wire, the diameter is $2d$.

On Fig. 6.2, sketch a graph to show the variation of the average drift speed with position along the wire between S and T.

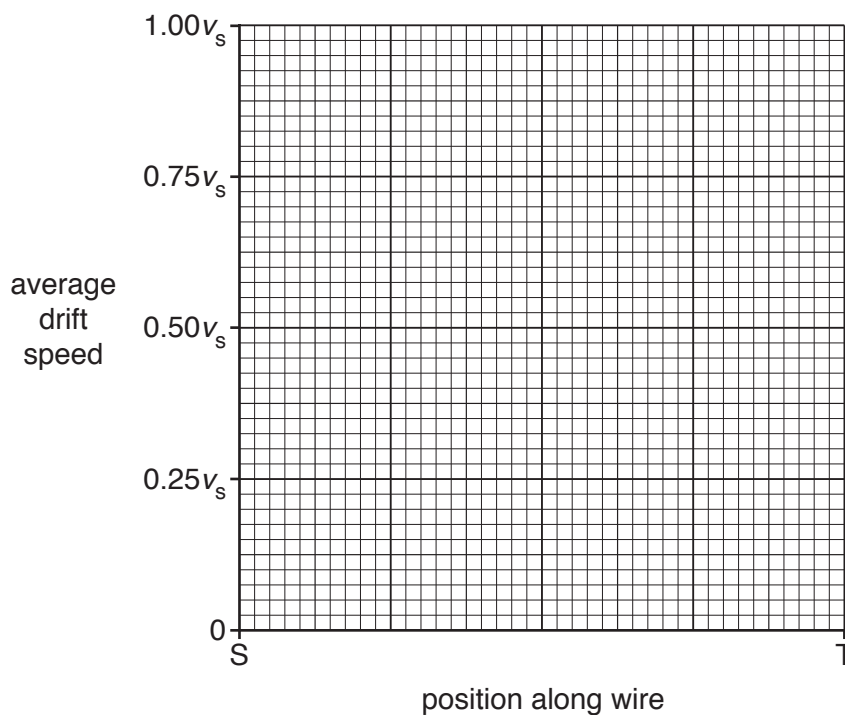


Fig. 6.2

[2]

24 (a) Define the *ohm*.

.....[1]

(b) A battery of electromotive force (e.m.f.) E and internal resistance 1.5Ω is connected to a network of resistors, as shown in Fig. 6.1.

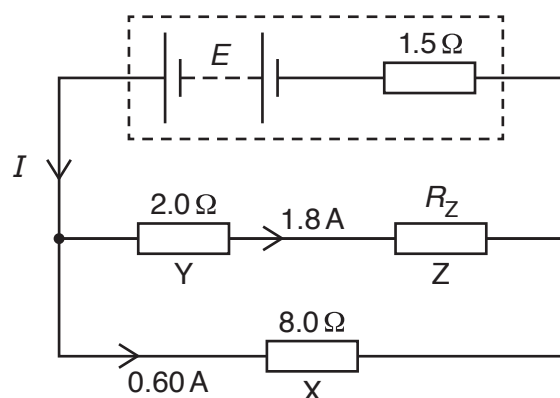


Fig. 6.1

Resistor X has a resistance of 8.0Ω . Resistor Y has a resistance of 2.0Ω . Resistor Z has a resistance of R_Z . The current in X is 0.60 A and the current in Y is 1.8 A .

(i) Calculate:

- the current I in the battery

$$I = \dots\dots\dots \text{ A [1]}$$

- resistance R_Z

$$R_Z = \dots\dots\dots \Omega [2]$$

- e.m.f. E .

$$E = \dots\dots\dots \text{ V [2]}$$

- (ii) Resistors X and Y are each made of wire. The two wires have the same length and are made of the same metal.

Determine the ratio:

1.
$$\frac{\text{cross-sectional area of wire X}}{\text{cross-sectional area of wire Y}}$$

ratio = [2]

2.
$$\frac{\text{average drift speed of free electrons in X}}{\text{average drift speed of free electrons in Y}}$$

ratio = [2]

25 The current I in a metal wire is given by the expression

$$I = Anve$$

where v is the average drift speed of the free electrons in the wire and e is the elementary charge.

(a) State what is meant by the symbols A and n .

A :

n :

[2]

(b) Use the above expression to determine the SI base units of e .
Show your working.

base units [2]

(c) Two lamps P and Q are connected in series to a battery, as shown in Fig. 6.1.

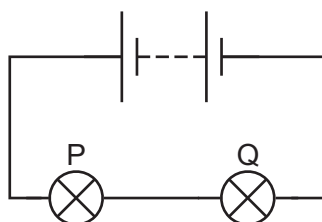


Fig. 6.1

The radius of the filament wire of lamp P is twice the radius of the filament wire of lamp Q.
The filament wires are made of metals with the same value of n .

Calculate the ratio

$$\frac{\text{average drift speed of free electrons in filament wire of P}}{\text{average drift speed of free electrons in filament wire of Q}}.$$

ratio = [2]

26. 9702/22/O/N/20/Q6

(a) A network of three resistors of resistances R_1 , R_2 and R_3 is shown in Fig. 6.1.

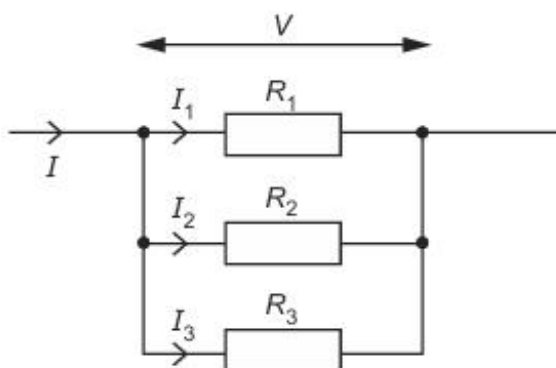


Fig. 6.1

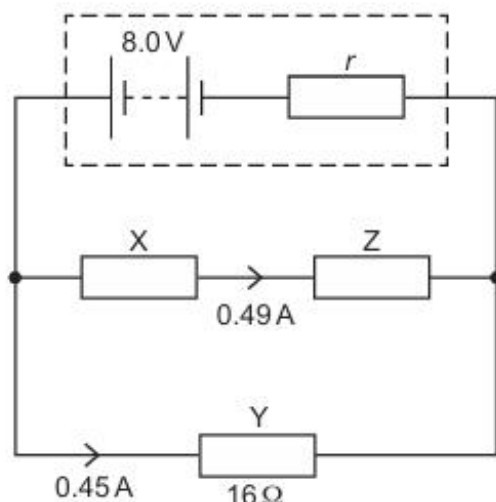
The individual currents in the resistors are I_1 , I_2 and I_3 . The total current in the combination of resistors is I and the potential difference across the combination is V .

Show that the combined resistance R of the network is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}.$$

[2]

(b) A battery of electromotive force (e.m.f.) 8.0V and internal resistance r is connected to three resistors X, Y and Z, as shown in Fig. 6.2.



Resistor Y has a resistance of $16\ \Omega$. The current in resistor X is 0.49 A and the current in resistor Y is 0.45 A .

Calculate:

- (i) the current in the battery

current = A [1]

- (ii) the internal resistance r of the battery.

$r =$ Ω [2]

- (c) Resistors X and Y in Fig. 6.2 are made from wires of the same material and cross-sectional area. The average drift speed of the free electrons in X is $2.1 \times 10^{-4}\text{ ms}^{-1}$.

Calculate the average drift speed v of the free electrons in Y.

$v =$ ms^{-1} [2]

- (d) Resistor Z in Fig. 6.2 is replaced by a new resistor of smaller resistance.

State and explain the effect, if any, on the terminal potential difference of the battery.

.....
.....
..... [2]

[Total: 9]

27. 9702/22/M/J/21/Q5

- (a) Define the *ohm*.

.....
..... [1]

- (b) A wire is made of metal of resistivity ρ . The length L of the wire is gradually increased. Assume that the volume V of the wire remains constant as its length is increased.

Show that the resistance R of the extending wire is proportional to L^2 .

[2]

- (c) A battery of electromotive force (e.m.f.) E and internal resistance r is connected to a variable resistor of resistance R , as shown in Fig. 5.1.

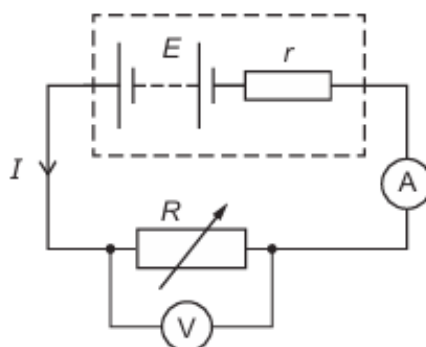


Fig. 5.1

An ammeter measures the current I in the circuit. A voltmeter measures the potential difference V across the variable resistor.

The resistance R is now varied to change the values of I and V .

The variation with I of V is shown in Fig. 5.2.

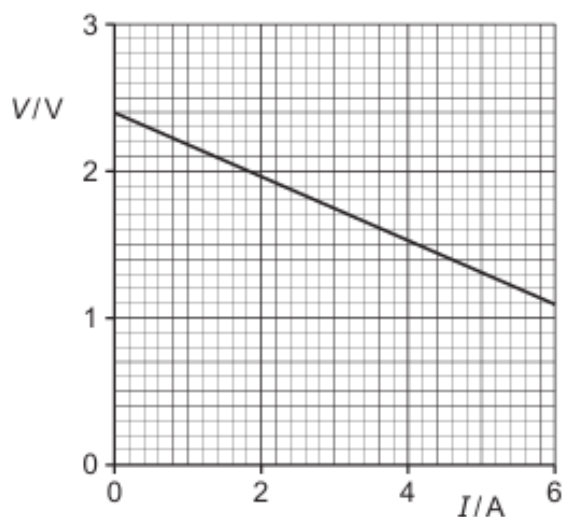


Fig. 5.2

- (i) Use Fig. 5.2 to state the e.m.f. E of the battery.

$E = \dots\dots\dots$ V [1]

- (ii) Use Fig. 5.2 to determine the power dissipated in the variable resistor when there is a current of 5.0 A.

power = $\dots\dots\dots$ W [3]

- (iii) State what is represented by the value of the gradient of the graph.

$\dots\dots\dots$ [1]

[Total: 8]

28. 9702/22/O/N/21/Q6

A cell of electromotive force (e.m.f.) 0.48 V is connected to a metal wire X, as shown in Fig. 6.1.

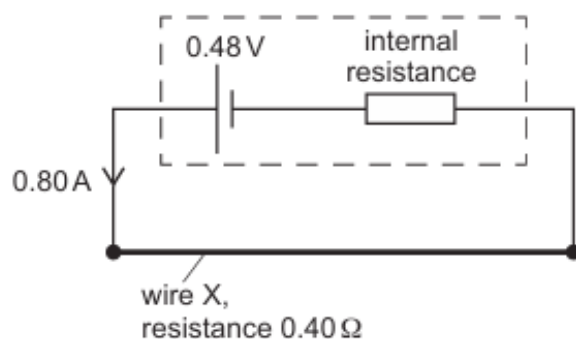


Fig. 6.1

The cell has internal resistance. The current in the cell is 0.80 A .

Wire X has length 3.0 m , cross-sectional area $1.3 \times 10^{-7}\text{ m}^2$ and resistance $0.40\ \Omega$.

(a) Calculate the charge passing through the cell in a time of 7.5 minutes .

charge = C [2]

(b) Calculate the percentage efficiency with which the cell supplies power to wire X.

efficiency = % [3]

- (c) There are 3.2×10^{22} free (conduction) electrons contained in the volume of wire X.

For wire X, calculate:

- (i) the number density n of the free electrons

$$n = \dots\dots\dots \text{m}^{-3} \quad [1]$$

- (ii) the average drift speed of the free electrons.

$$\text{average drift speed} = \dots\dots\dots \text{ms}^{-1} \quad [2]$$

- (d) A wire Y has the same cross-sectional area as wire X and is made of the same metal. Wire Y is longer than wire X.

Wire X in the circuit is now replaced by wire Y. Assume that wire Y has the same temperature as wire X.

State and explain whether the average drift speed of the free electrons in wire Y is greater than, the same as, or less than that in wire X.

.....

.....

.....

.....

.....

..... [3]

[Total: 11]

29. 9702/22/F/M/22/Q6

The ends of a metal resistance wire are connected to a battery of electromotive force (e.m.f.) 8.0 V and negligible internal resistance, as shown in Fig. 6.1.

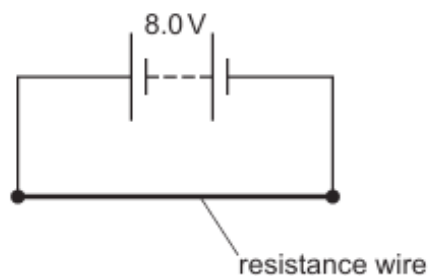


Fig. 6.1

The power dissipated by the resistance wire is 36 W .

(a) Calculate:

(i) the current in the resistance wire

current =A [2]

(ii) the number of free electrons that pass through the resistance wire in a time of 50 s

number = [2]

(iii) the resistance of the wire.

resistance = Ω [2]

- (b) The metal of the resistance wire in the circuit has a resistivity of $1.4 \times 10^{-6} \Omega \text{m}$. The cross-sectional area of the wire is 0.25 mm^2 .

Determine the length of the wire.

length =m [2]

- (c) The circuit shown in Fig. 6.1 is modified by replacing the original resistance wire with a second resistance wire. The second wire has a greater diameter than the original wire. There are no other differences between the second wire and the original wire.

By reference to resistance, state and explain whether the power dissipated by the second wire is more than, less than or the same as the power dissipated by the original wire.

.....
.....
.....
..... [2]

30. 9702/22/M/J/22/Q6

- (a) A network of three resistors of resistances R_1 , R_2 and R_3 is shown in Fig. 6.1.

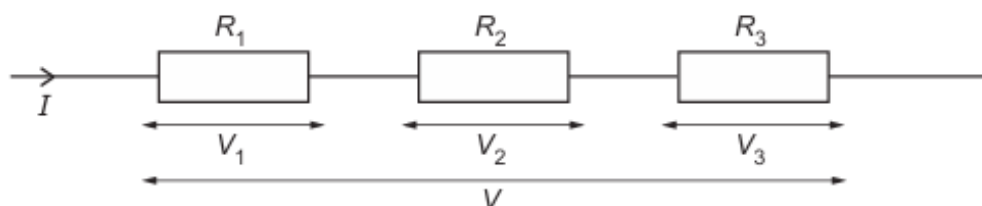


Fig. 6.1

The individual potential differences across the resistors are V_1 , V_2 and V_3 . The current in the combination of resistors is I and the total potential difference across the combination is V .

Show that the combined resistance R of the network is given by

$$R = R_1 + R_2 + R_3.$$

[2]

- (b) A battery of electromotive force (e.m.f.) 8.0V and negligible internal resistance is connected to a thermistor, a switch X and two fixed resistors, as shown in Fig. 6.2.

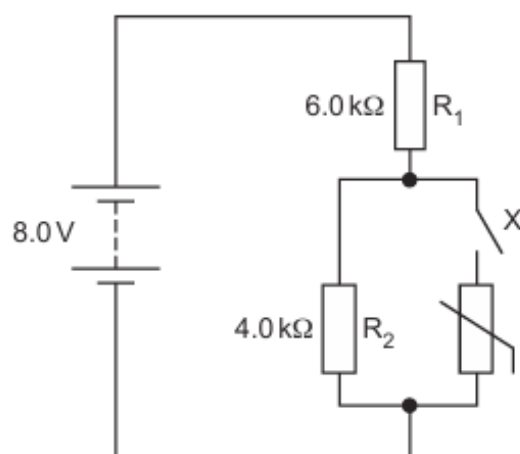


Fig. 6.2

Resistor R_1 has resistance 6.0 k Ω and resistor R_2 has resistance 4.0 k Ω .

- (i) Switch X is open.

Calculate the potential difference across R_1 .

potential difference = V [2]

- (ii) Switch X is now closed. The resistance of the thermistor is $12.0\text{ k}\Omega$.

Calculate the current in the battery.

current = A [2]

- (c) The switch X in the circuit in (b) remains closed. The temperature of the thermistor decreases.

By reference to the current in the battery, state and explain the effect, if any, of the decrease in temperature on the power produced by the battery.

.....
.....
.....
.....
..... [3]

[Total: 9]

31. 9702/23/M/J/22/Q6

(a) (i) On Fig. 6.1, sketch the I – V characteristic of a filament lamp.

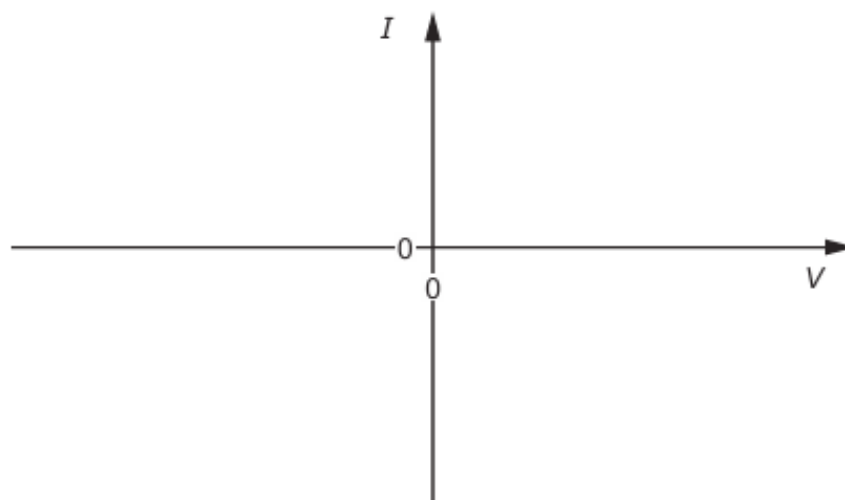


Fig. 6.1

[2]

(ii) Explain the shape of the line in (a)(i).

.....

.....

.....

..... [3]

(b) A conducting wire has length 5.8 m and cross-sectional area $3.4 \times 10^{-8} \text{ m}^2$. The resistivity of the metal of the wire is $5.6 \times 10^{-8} \Omega \text{ m}$.

Calculate the resistance of the wire.

resistance = Ω [2]

- (c) A resistor of resistance R is placed in a circuit with a cell of negligible internal resistance, two switches S_1 and S_2 , a second resistor of resistance $2R$ and three ammeters X, Y and Z. The circuit is shown in Fig. 6.2.

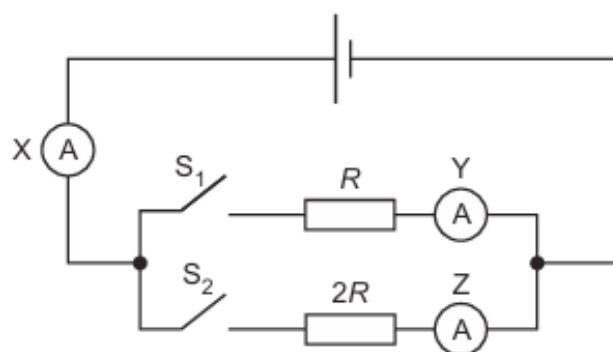


Fig. 6.2

The reading on X is 1.0A when S_1 is open and S_2 is closed.

Complete Table 6.1.

Table 6.1

position of switches		ammeter readings		
S_1	S_2	reading on X/A	reading on Y/A	reading on Z/A
open	open	0	0	0
open	closed	1.0		
closed	open			
closed	closed			

[4]

[Total: 11]

32. 9702/22/O/N/22/Q6

(a) Define electric potential difference.

.....
 [1]

(b) A battery is connected to two resistors X and Y, as shown in Fig. 6.1.

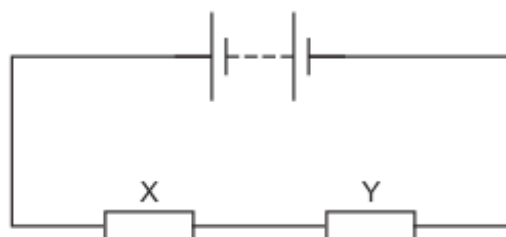


Fig. 6.1

The resistance of resistor X is greater than the resistance of resistor Y.

State and explain which resistor dissipates more power.

.....

 [3]

(c) A battery of electromotive force (e.m.f.) 9.0V and internal resistance r is connected to two resistors P and Q, as shown in Fig. 6.2.

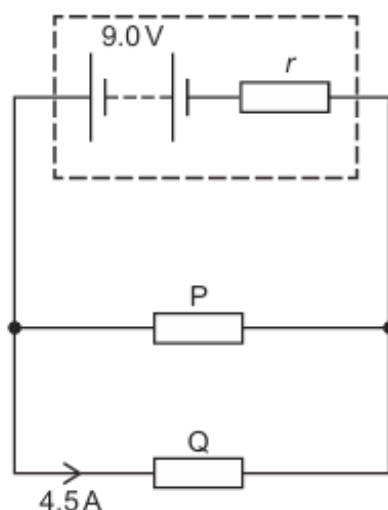


Fig. 6.2

A total charge of 650 C moves through resistor P in a time interval of 540 s. During this time resistor P dissipates 4800 J of energy. The current in resistor Q is 4.5 A. Assume that the e.m.f. of the battery remains constant.

Calculate:

- (i) the current in resistor P

current = A [2]

- (ii) the potential difference across resistor P

potential difference = V [2]

- (iii) the internal resistance r of the battery.

$r =$ Ω [2]

[Total: 10]

33. 9702/22/F/M/23/Q6

(a) Define the potential difference across a component.

.....
..... [1]

(b) The variation with potential difference V of the current I in a semiconductor diode is shown in Fig. 6.1.

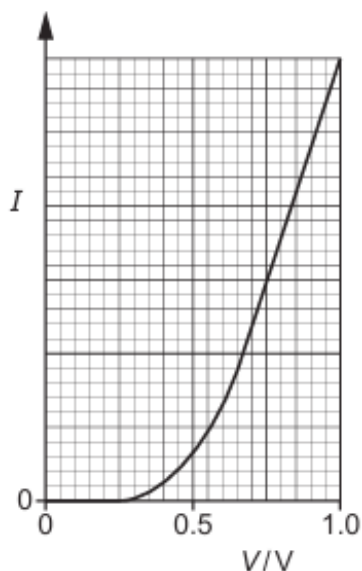


Fig. 6.1

Use Fig. 6.1 to describe qualitatively:

(i) the resistance of the diode in the range $V = 0$ to $V = 0.25\text{ V}$

..... [1]

(ii) the variation, if any, in the resistance of the diode as V changes from $V = 0.75\text{ V}$ to $V = 1.0\text{ V}$.

..... [1]

- (c) A battery of electromotive force (e.m.f.) 12 V and negligible internal resistance is connected to a uniform resistance wire XY, a fixed resistor and a variable resistor, as shown in Fig. 6.2.

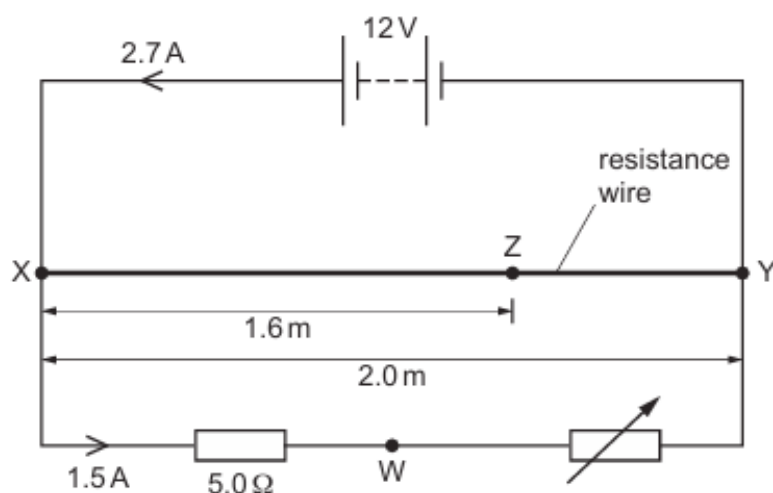


Fig. 6.2 (not to scale)

The fixed resistor has a resistance of $5.0\ \Omega$. The current in the battery is 2.7 A and the current in the fixed resistor is 1.5 A.

- (i) Calculate the current in the resistance wire.

current = A [1]

- (ii) Determine the resistance of the variable resistor.

resistance = Ω [2]

- (iii) Wire XY has a length of 2.0 m. Point Z on the wire is a distance of 1.6 m from point X. The fixed resistor is connected to the variable resistor at point W.

Determine the potential difference between points W and Z.

potential difference = V [3]

- (iv) The resistance of the variable resistor is now increased.

By considering the currents in every part of the circuit, state and explain whether the total power produced by the battery decreases, increases or stays the same.

.....
.....
.....
.....
.....
..... [3]

[Total: 12]

34. 9702/21/M/J/23/Q6

A metal wire in a circuit has a length of 1.8m and a cross-sectional area of $1.5 \times 10^{-6} \text{ m}^2$.
The total number of free electrons (charge carriers) in the wire is 2.3×10^{23} .

There is a current in the wire so that a charge of 172C moves past a fixed point in the wire in a time of 2.5 minutes.

(a) Show that the number density of the free electrons in the wire is $8.5 \times 10^{28} \text{ m}^{-3}$.

[1]

(b) Calculate the average drift speed of the free electrons.

average drift speed = ms^{-1} [3]

[Total: 4]

35. 9702/22/M/J/23/Q6

- (a) The current in a filament lamp decreases.

State and explain how the resistance of the lamp changes.

.....
..... [1]

- (b) A cylindrical wire has length L and resistance R . The **total** number of free electrons (charge carriers) contained in the volume of the wire is N . Each free electron has charge e . The potential difference between the ends of the wire is V .

Determine expressions, in terms of some or all of the symbols e , L , N , R and V for:

- (i) the current in the wire

current = [1]

- (ii) the average drift speed of the free electrons

average drift speed = [2]

- (iii) the average time taken for a free electron to move along the full length of the wire.

time taken = [1]

[Total: 5]

36. 9702/23/M/J/23/Q5

A student sets up a circuit with a battery, an ammeter, a heater and a light-dependent resistor (LDR) all in series.

The battery has negligible internal resistance.

A voltmeter is connected across (in parallel with) the heater.

(a) On Fig. 5.1, complete the circuit diagram of this arrangement.



Fig. 5.1

[3]

(b) The heater is a wire made of metal of resistivity $1.1 \times 10^{-6} \Omega \text{ m}$. The wire has length 2.0 m and cross-sectional area $3.8 \times 10^{-7} \text{ m}^2$.

The reading on the voltmeter is 4.8 V.

Calculate:

(i) the resistance of the heater

resistance = Ω [2]

(ii) the reading on the ammeter.

reading on ammeter = A [1]

- (c) The heater is replaced by a new wire. The new wire is made of the same metal as the wire in (b) and has the same length but a larger diameter.

The resistance of the LDR remains constant.

- (i) State and explain whether the new wire has a resistance that is greater than, less than or the same as that of the wire in (b).

.....
.....
..... [2]

- (ii) State and explain whether the new reading on the voltmeter is greater than, less than or equal to 4.8 V.

.....
.....
..... [2]

[Total: 10]

37. 9702/21/O/N/23/Q6

A battery is connected in a circuit with a light-dependent resistor (LDR), two fixed resistors and a voltmeter, as shown in Fig. 6.1.

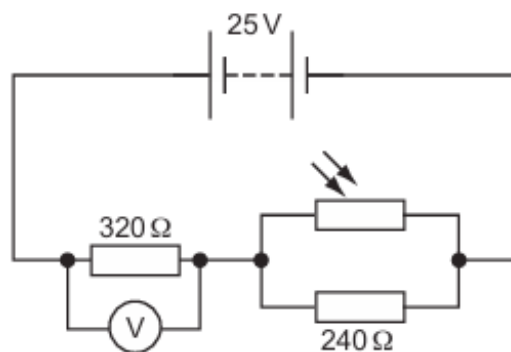


Fig. 6.1

The battery has an electromotive force (e.m.f.) of 25 V and negligible internal resistance. The resistors have resistances of 320 Ω and 240 Ω .

- (a) The voltmeter displays a reading of 16 V.
- (i) Show that the current in the battery is 0.050 A.

[1]

- (ii) Calculate the resistance of the LDR.

resistance = Ω [3]

(iii) Determine the ratio

$$\frac{\text{power dissipated in the LDR}}{\text{power dissipated in the } 240\,\Omega \text{ resistor}}$$

ratio = [2]

(b) The intensity of the light incident on the LDR increases.

State and explain what happens to the voltmeter reading.

.....
.....
.....
..... [3]

[Total: 9]

38. 9702/22/O/N/23/Q6

- (a) A metal wire has a resistance per unit length of $0.92\Omega\text{m}^{-1}$. The wire has a uniform cross-sectional area of $5.3 \times 10^{-7}\text{m}^2$.

Calculate the resistivity of the metal of the wire.

resistivity = Ωm [2]

- (b) A battery of electromotive force (e.m.f.) E and negligible internal resistance is connected in series with a fixed resistor and a light-dependent resistor (LDR), as shown in Fig. 6.1.

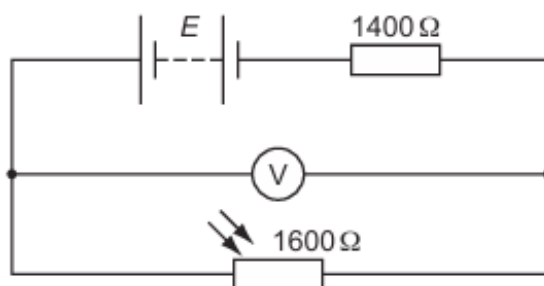


Fig. 6.1

The resistance of the fixed resistor is 1400Ω . The intensity of the light illuminating the LDR causes it to have a resistance of 1600Ω . A voltmeter connected across the LDR reads 6.4V .

- (i) Show that the current in the LDR is $4.0 \times 10^{-3}\text{A}$.

[1]

- (ii) Calculate the number of free electrons passing through the LDR in a time of 3.2 minutes.

number of free electrons = [2]

(iii) Calculate the e.m.f. E .

$$E = \dots\dots\dots \text{ V [2]}$$

(iv) Determine the ratio

$$\frac{\text{power dissipated in LDR}}{\text{power dissipated in fixed resistor}}.$$

$$\text{ratio} = \dots\dots\dots \text{ [2]}$$

(c) The environmental conditions change causing a decrease in the resistance of the LDR in (b). The temperature of the environment does not change.

State whether there is a decrease, increase or no change to:

(i) the intensity of the light illuminating the LDR

..... [1]

(ii) the current in the battery

..... [1]

(iii) the reading of the voltmeter.

..... [1]

[Total: 12]

39. 9702/23/O/N/23/Q7

(a) State Ohm's law.

.....

 [2]

(b) A battery of electromotive force (e.m.f.) 6.2V and negligible internal resistance is connected in a circuit to a uniform resistance wire, a voltmeter, a fixed resistor and a switch, as shown in Fig. 7.1.

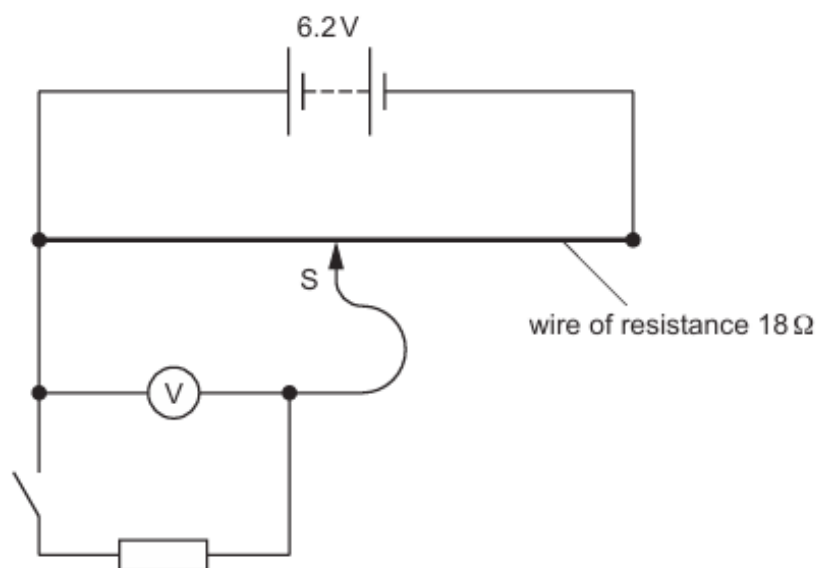


Fig. 7.1

The resistance wire has resistance 18Ω , length 0.94 m and cross-sectional area $7.2 \times 10^{-8}\text{ m}^2$. The slider S is positioned half-way along the length of the wire.

(i) Calculate the resistivity ρ of the material of the resistance wire.

$\rho = \dots\dots\dots \Omega\text{m}$ [2]

(ii) The switch is open.
 State the reading on the voltmeter.

voltmeter reading = $\dots\dots\dots \text{V}$ [1]

(iii) The switch is now closed.

State whether there is an increase, decrease or no change to:

- the current in the battery

.....

- the voltmeter reading.

.....

[2]

(iv) The switch remains closed. The slider S is moved along the resistance wire so that the voltmeter reading is 3.1 V.

On Fig. 7.1, draw a cross (×) on the resistance wire to show a possible new position of the slider. [1]

(c) The circuit in (b) is altered by changing the battery for one of a different e.m.f. The switch is open.

A student records the following data for the resistance wire:

current in the wire	0.93 A
mean drift speed of charge carriers	$1.3 \times 10^{-3} \text{ m s}^{-1}$
number density of charge carriers	$9.0 \times 10^{28} \text{ m}^{-3}$

(i) Determine the charge q of a charge carrier in the wire suggested by this data.

$q = \dots\dots\dots$ C [2]

(ii) With reference to the value of q , explain why the data recorded by the student cannot be correct.

.....

..... [1]

[Total: 11]